

Introduction to Data Structure and Algorithm I

Liting Zhang

University of West Florida

Course and Instructor Information

- Class schedule: T, R 9:30 - 10: 45 AM
- Meets: 4/248
- Instructor: Liting Zhang, PhD
- Office Hours:
 - ▶ Office: 4/435
 - ▶ Tue and Thurs: 2 - 4 pm
 - ▶ zoom <https://uwf.zoom.us/j/9979287260>
- Email: lzhang@uwf.edu

Communication Policy

- Course information and updates are sent via Canvas announcements
- Email is the preferred method of communication
- Use subject line format: COP3530: [Your Topic Here]
- Example: COP3530: Project 2 Questions
- Expect a response within 1–2 business days
- Avoid:
 - ▶ Canvas/Navigation messages
 - ▶ Sending only error screenshots without your code/context

Important Note

- This is a programming-intensive course.
- It does **not** get easier as the semester goes on:
 - ▶ you will read code, debug, and write substantial programs
- This is **not** self-paced:
 - ▶ complete assignments and programs by the posted deadlines

Course Objectives

- Develop solutions by applying presented algorithms
- Write algorithms for solving problems prior to coding
- Analyze solutions in terms of time and space complexity
- Develop programs that utilize different types of data structures
- Write recursive definitions and develop recursive programming solutions

Textbook Info (zyBooks)

- Go to zyBooks website and create a new account
- Enter zyBook code: UWFCOP3530ZhangSpring2026
- Subscribe
- Participation activities: do before/during reading
- Challenges: do after covered in lecture
- Grading: finish $> 80\%$ for full credit; graded after due date; counts toward final grade

Attendance Policy

- Regular attendance is expected and will greatly improve performance
- Attendance is not taken and not part of your grade
- In-class activities are graded (lowest two will be dropped)

Grading / Evaluation

Assignment	% of Grade
Checkpoints	10%
Projects	30%
Tests	20%
Final Exam	20%
zyBooks Activities	10%
Classroom Activities	10%

Letter Grade	Range
A	92% – 100%
A-	90% – 91.9%
B+	87% – 89.9%
B	84% – 86.9%
B-	80% – 83.9%
C+	77% – 79.9%
C	74% – 76.9%
C-	70% – 73.9%
D+	67% – 69.9%
D	60% – 66.9%
F	0% – 59.9%

Assessments

- Checkpoints:
 - ▶ 2 mini programming assignments (GitHub Classroom)
 - ▶ about 2 weeks to finish
- Projects:
 - ▶ 3 programming assignments (GitHub Classroom)
 - ▶ about 3 weeks to finish
- Exams:
 - ▶ 2 tests (around week 6 and week 11)
 - ▶ 1 comprehensive final exam
 - ▶ Respondus LockDown Browser required
- Activities:
 - ▶ zyBooks activities
 - ▶ classroom activities
 - ▶ short review quiz for previous class

Late Policy

- In-class activities: no late submissions
- zyBooks: due as shown in Canvas; no late submissions
- Checkpoints/Projects:
 - ▶ accepted up to 48 hours after deadline; after that, 0 without approved extension
 - ▶ no late submission for the last project
 - ▶ penalty (applied once to earned score):
 - ★ ≤ 24 hours late: -10%
 - ★ 24–48 hours late: -20%
 - ▶ extension: request before the deadline when possible

Excused Absences Policy

- Accepted reasons (with documentation):
 - ▶ illness, family death/crisis
 - ▶ military/jury duty, religious holy days
 - ▶ official University events
 - ▶ serious illness of dependent children
- Notify instructor ASAP for any excused absence affecting graded work
- Make-up tests:
 - ▶ allowed with medical notice or approved emergency
 - ▶ non-medical/emergency reasons require advance approval
 - ▶ if absent without prior approval, notify within 48 hours
 - ▶ make up within 2 weeks (unless long-term illness)

Development Environment for C++

- Minimal requirements
- Toolchain and environment
- Command-line environment (bash)
- Local development environment
 - ▶ Linux native
 - ▶ Windows WSL (Linux CLI)
 - ▶ Mac OS native
- Remote development environment (optional)
 - ▶ SSH server (Linux CLI)
 - ▶ GitHub Codespaces (Web IDE + Linux VM)
- Traps and pitfalls
 - ▶ MinGW / MSYS2 / Git Bash
 - ▶ click-and-run single-file workflows

Minimal Requirements for C++ Development

- An editor/IDE to edit source code files
 - ▶ Visual Studio Code (recommended)
- A build environment to compile source files into executables/libraries
 - ▶ access through a command-line interface (CLI), usually bash
 - ▶ core software: g++, make, git, valgrind
- Other bash tools may be useful (zip, tar, grep, etc.)

Windows Users (Suggested Workflow)

- NEVER put your project in OneDrive folder
- NEVER put your project under a directory containing spaces in its name
- AVOID MinGW/MSYS2/Git Bash
- Pay attention to line endings for files obtained from Windows apps
- NEVER download and unzip from GitHub
 - ▶ use `git clone` instead (zip tools often mess up line endings)

Tools and Environment

- `make`

- ▶ automate the build process and manage dependencies
- ▶ read more about `make`

- `git`

- ▶ distributed version control for text files (source code)
- ▶ we use a small subset as beginners
- ▶ backup versions of your work; submit your work
- ▶ read more about `git`

Command-Line Interface (CLI)

- Command-line interface (CLI): text-based interface to interact with the computer
- Shell: interprets commands
- Terminal: displays input/output
- Counterpart of GUI (graphical user interface)
- Why CLI:
 - ▶ Flexible: adjust parameters on the fly
 - ▶ Easy-to-use: easily automated even when complicated
 - ▶ Portable: same commands across platforms
 - ▶ Efficient: less resource consumption

Problems with GUI

- GUI is great, but not suitable to our needs
- Hard to automate
- Hard to learn well for modular projects + build systems

Local Environment

- A local development environment is recommended for this course
- Not always the most convenient, but worth your time to learn

Local Environment: Linux machine

- Install g++ and make using your package manager (Ubuntu example):

```
1 | sudo apt update
2 | sudo apt install build-essential
3 | # optional but recommended:
4 | sudo apt install valgrind
```

For other distributions, use the appropriate package manager.

Local Environment: Windows

- Install Ubuntu 20.04 LTS from Microsoft Store (WSL)
- Toolchain setup is similar to Linux

```
1 | sudo apt update
2 | sudo apt install build-essential
3 | # optional but recommended:
4 | sudo apt install valgrind
```

- Install Visual Studio Code
- Use VS Code + WSL workflow
- Demo: Week 1 WSL + VS Code (Classroom computer demo)

Local Environment: Mac OS

- Install Xcode Command Line Tools

```
1 | xcode-select --install
```

- Note: valgrind is not directly available on Mac OS

Remote Environment

- Remote environment: use a remote computer to do development
- SSH server (from CS department):
 - ▶ C++ toolchain pre-installed
 - ▶ handle files remotely
 - ▶ best with VS Code in SSH remote mode
- GitHub Codespaces:
 - ▶ in-browser VS Code + Linux virtual machine
 - ▶ free hours for GitHub Classroom

Traps and Pitfalls

- MinGW/MSYS2/Git Bash:
 - ▶ Windows port of g++ toolchain
 - ▶ g++ under MinGW may not match Linux behavior
 - ▶ compatibility issues with make
 - ▶ not recommended for this course
- Click-and-run features:
 - ▶ tricky to configure for modular projects with make
 - ▶ command line is usually better
- Examples: jGrasp, “Run and Debug” button in VS Code, OnlineGDB, Eclipse

GitHub and GitHub Classroom: why we use them

- **Git** is version control: track changes, revert mistakes, collaborate
- **GitHub** hosts git repositories online (backup + sharing)
- **GitHub Classroom** distributes assignments as private repos
- You submit by **pushing commits** (no file uploads, no zip submissions)

Key idea: repository = your assignment folder

- Each assignment link creates a **private repository** for you
- You clone it once, then:
 - ▶ edit code locally
 - ▶ commit changes
 - ▶ push to GitHub (this is the submission)
- Your repo keeps history: you can see what changed and when

Vocabulary you must know

- **repo** (repository): project folder tracked by git
- **clone**: download a repo to your machine (first time)
- **commit**: save a checkpoint with a message
- **push**: upload commits to GitHub (submission happens here)
- **pull**: download updates from GitHub (rare for assignments)
- **origin**: default remote repo on GitHub

Workflow overview (every assignment)

- 1 Accept assignment link (GitHub Classroom)
- 2 Clone repository to your machine (once)
- 3 Edit and test locally
- 4 Commit frequently with clear messages
- 5 Push to GitHub before the deadline

Step 1: accept assignment + clone

After accepting the [lab1 link](#), copy the repo URL, then:

```
1 # choose a folder where you store class work
2 mkdir -p ~/Desktop/cop3530
3 cd ~/Desktop/cop3530
4
5 # clone your private assignment repository
6 git clone your_private_git_repo_link
7 cd assignment-1-yourname
```

- **Do not** download ZIP files from GitHub.
- Clone once; after that you just work inside the folder.

Step 2: edit, build, run

Typical local cycle:

```
1  # see what files exist
2  ls
3  # edit main.cpp
4  # build (example)
5  make
6
7  # run (example)
8  ./main
```

Always make sure your code compiles and runs **before** you push.

Step 3: check status + add files

```
1  # see what changed  
2  git status  
3  
4  # stage changes (most common)  
5  git add .  
6  
7  # or stage a specific file  
8  git add main.cpp
```

- `git status` tells you exactly what will be committed.
- `git add .` stages all changes in the repo folder.

Step 4: commit with a message

```
1 | git commit -m "Implement insert() and fix edge cases"
```

Good commit messages:

- describe **what** changed (and why, if needed)
- are short and specific
- happen often (small checkpoints)

Step 5: push (this is submission)

```
1 | git push
```

- If this command succeeds, your code is on GitHub.
- GitHub Classroom autograding (if enabled) runs after pushes.
- Always push **before** the deadline.

How to confirm your submission

- Open your assignment repo on GitHub
- Check the README grade badge
- View grading detail
 - ▶ On your GitHub repo page, Click the Actions tab to see your graded results.
 - ▶ If it isn't a green check mark  patthen at least part of the testing failed.
 - ▶ Click the commit message for the failing version then click "Autograding" on the left side of the page.
 - ▶ Follow the  path and expand things to see what errors exist.
 - ▶ It is usually education/autograding v1 and you can expend this path to view the detail.

Common mistakes (avoid these)

- Editing files outside the repo folder (then nothing gets committed)
- Forgetting `git add` (commit contains no changes)
- Forgetting `git push` (GitHub does not receive your commits)
- Downloading ZIP from GitHub (breaks workflow + line endings)
- Working in OneDrive / folders with spaces (Windows issues)

When things go wrong: quick fixes

- “I committed but GitHub doesn’t update”:
1 | `git push`
- “Nothing to commit, but I changed files”:
1 | `git status`
2 | `git add .`
3 | `git commit -m "Update solution"`
- “Merge / pull issues” (rare for assignments):
 - ▶ ask instructor/TA before force commands

Recommended habit: commit early, commit often

- Treat commits like save points
- Make a commit when you:
 - ▶ pass a test
 - ▶ finish a function
 - ▶ fix a bug
- If something breaks, you can go back to a working commit

Comment and Documentation

- Comment

- ▶ audience: the programmer themselves
- ▶ used to explain code to yourself/other programmers
- ▶ often replaceable by good names and good structure

- Documentation

- ▶ audience: users of the code (including future you)
- ▶ explains behavior, contracts, parameters, edge cases
- ▶ always useful

Comment vs Documentation (examples)

```
1 // initialize the array (often replaceable by a good function name)
2 for (int i = 0; i < 10; ++i) {
3     a[i] = 0;
4 }
5
6 // print the array
7 for (int i = 0; i < 10; ++i) {
8     std::cout << a[i] << std::endl;
9 }
10
11 /**
12  * @brief Initialize an array to all zeros.
13  * @param a The array to initialize.
14  * @param n The number of elements in the array.
15  */
16 void initializeArray(int a[], int n);
17
18 void initializeArray(int a[], int n) {
19     for (int i = 0; i < n; ++i) {
20         a[i] = 0;
21     }
22 }
```

Types of Errors in C++

- Syntax error
- Runtime error
- Logic error

Syntax Error

- Compilation error: compiler cannot understand your code
- Reading error messages helps, but can be non-trivial at first
- A good editor/IDE can highlight syntax errors (may require configuration)

Runtime Error

- Related concept: exceptions
- Compiler understands your code, but the program fails while running
- Symptoms: crash, segmentation fault, freeze, etc.

Logic Error

- Program compiles and runs, but does not do what you want
- Problems it causes:
 - ▶ wrong output / wrong behavior
 - ▶ wasted resources (too slow; too much memory; possible memory leak)
- Often the most difficult type of error to debug

Introduction to Data Structures and Algorithms

1 Introduction to Data Structures and Algorithms

2 Abstract Data Types (ADTs)

3 Review

Data Structures

- A data structure is a way of organizing, storing, and performing operations on data.

Overview

- Data structure
- Introduction to algorithms
- Relation between data structure and algorithms
- Abstract data types
- Applications of ADTs

Data structures: four common examples

- **Record:** group related fields (e.g., Student: id, name, age)
- **Array:** contiguous sequence (e.g., test scores)
- **Linked list:** nodes connected by pointers (e.g., train cars)
- **Binary tree:** hierarchical structure (e.g., directory tree)

Example: Record and Array

Record (Student)

- id: 1234567
- name: Alice
- age: 20

Why it helps: keeps related data together.

Array (Scores)

- contiguous memory
- constant-time indexing

78	90	92	100	85	88
----	----	----	-----	----	----

Example: Linked list and Binary tree

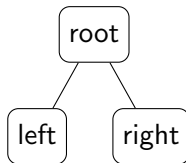
Linked list

- nodes store data + link to next
- flexible insert/remove



Binary tree

- hierarchical relationships
- supports efficient search (when structured)



Algorithms

- An algorithm is a step-by-step process to solve a problem.
- Input $\rightarrow$ process $\rightarrow$ output
- Same problem can have many algorithms with different efficiency.

Algorithms + data structures: together

- Data structure organizes data; algorithm uses that organization.
- Examples:
 - ▶ Searching a sorted array: **binary search**
 - ▶ Navigation / shortest path: **Dijkstra's algorithm** on a **graph**

Example problem: find maximum value

- Problem: given an array of numbers, return the maximum element.
- Idea: scan from left to right while tracking the best so far.

Why this matters:

- establishes a baseline linear-time algorithm: $O(n)$
- helps compare to slower approaches (e.g., nested loops)

Find maximum (C++ example)

```
1 int maxValue(const int a[], int n) {  
2     int best = a[0];  
3     for (int i = 1; i < n; ++i) {  
4         if (a[i] > best) best = a[i];  
5     }  
6     return best;  
7 }
```

Runs in $O(n)$ time and $O(1)$ extra space.

Practical Applications of Algorithms

- Searching (databases, files, websites)
- Sorting (ranking, reporting, preprocessing)
- Graph algorithms (routing, networks, recommendations)
- Compression and hashing (storage, security, performance)
- Optimization (scheduling, planning, resource allocation)

Example Computational Problems

- Find an item quickly in a large dataset
- Plan the shortest/cheapest route between locations
- Schedule tasks with priorities and deadlines
- Match or recommend items based on similarity
- Detect duplicates, anomalies, or patterns

Efficient Algorithms and Hard Problems

- Efficient algorithms: runtime increases no more than polynomially with input size
- Some problems lack known efficient algorithms → NP-complete problems
- NP-complete problem characteristics:
 - ▶ no efficient algorithm found
 - ▶ no proof that an efficient algorithm is impossible
 - ▶ if one NP-complete problem is solved efficiently, all can be
- Strategy:
 - ▶ focus on finding good (not necessarily optimal) solutions

Abstract Data Types (ADTs)

- 1 Introduction to Data Structures and Algorithms
- 2 Abstract Data Types (ADTs)
- 3 Review

Abstract Data Types (ADTs)

- An ADT specifies **what operations** are supported (behavior), not **how** they are implemented.
- Focus: interface, contracts, and expected behavior.
- Multiple implementations can satisfy the same ADT.

Common ADTs

- List
- Stack
- Queue / Deque
- Priority queue
- Set / Map (dictionary)
- Graph

Application of ADTs

- Choose an ADT based on what operations you need:
 - ▶ stack: last-in-first-out
 - ▶ queue: first-in-first-out
 - ▶ map: key $\rightarrow$ value lookup
 - ▶ priority queue: always remove highest priority
- Then choose an implementation based on performance goals:
 - ▶ array vs linked list
 - ▶ binary heap vs balanced tree
 - ▶ hash table vs tree-based map

ADTs in Standard Libraries

- C++ STL examples:
 - ▶ `std::vector` (dynamic array)
 - ▶ `std::list` (doubly linked list)
 - ▶ `std::stack`, `std::queue`, `std::priority_queue`
 - ▶ `std::map`, `std::unordered_map`
 - ▶ `std::set`, `std::unordered_set`
- These are implementations that provide ADT-like interfaces

Review

- 1 Introduction to Data Structures and Algorithms
- 2 Abstract Data Types (ADTs)
- 3 Review

Summary

- Policies + workflow matter: they prevent avoidable problems
- CLI + Linux toolchain is the most reliable setup for this course
- Classify errors first (syntax/runtime/logic), then debug efficiently
- Data structures organize data; algorithms solve problems using that organization
- ADTs describe behavior; implementations determine performance